

FND-A Foundations: sets, maps, induction, and proof patterns

Definition 1. For sets A, B , a function $f: A \rightarrow B$ assigns each $a \in A$ exactly one value $f(a) \in B$. It is injective if $f(a) = f(b) \implies a = b$, surjective if $\forall b \in B \exists a \in A: f(a) = b$, and bijective if both.

Theorem 2 [FND-A] (inverse criterion). A map $f: A \rightarrow B$ is bijective iff there exists $g: B \rightarrow A$ with $g \circ f = \text{id}_A$ and $f \circ g = \text{id}_B$. The two identities are both required unless injectivity or surjectivity is already $P(n)$, contradiction.
Proof. If f is bijective, define $g(b)$ as the unique preimage. Conversely, $g \circ f = \text{id}$ implies injectivity: $f(a) = f(a') \implies a = g(f(a)) = g(f(a')) = a'$. The other identity gives surjectivity since $b = f(g(b))$.

Assumption. Unless stated otherwise, families indexed by $\mathbb{N}$ are assumed to have unique $\mathbb{N}$ -indexed elements to the ambient universe, empty sum $\emptyset$, and empty product 1 .
De Morgan: $(\bigcup_i A_i)^c = \bigcap_i A_i^c$ and $(\bigcap_i A_i)^c = \bigcup_i A_i^c$. For predicates P, Q , $\neg(P \wedge Q) \equiv \neg P \vee \neg Q$ and $\neg(P \vee Q) \equiv \neg P \wedge \neg Q$. Negating quantifiers swaps them: $\neg(\forall x P(x)) \equiv \exists x \neg P(x)$, while $\neg(\exists x P(x)) \equiv \forall x \neg P(x)$. A single quantifier over a finite domain is finite; examples cannot prove it.

Induction and external arguments

Principle 3. To prove $P(n)$ for all $n \geq n_0$, prove $P(n_0)$ and $P(k) \implies P(k+1)$ for arbitrary $k \geq n_0$. Strong induction may assume $P(n_0), \dots, P(k)$; it is logically equivalent.
Example 1. Show $1 + 2 + \dots + (2n-1) = n^2$. The base case is $1 = 1^2$. If the claim holds for $n = k$, then adding $2k+1$ gives $k^2 + 2k + 1 = (k+1)^2$. State the arbitrary k and close the quantifier.
Caution. Do not assume the desired statement at $k+1$, and do not hide a stronger induction hypothesis. For recurrences, check the base case. An extremal proof selects a smallest or largest counterexample. Verify that the candidate set is nonempty and that the order is well founded. Minimal-counterexample template: assume failure; choose least m failing; construct a smaller failing value n ; contradiction.
gwi [FND-A] equivalence relations. Let $x \sim y$ iff $x, y \in \mathbb{Z}$ on $\mathbb{R}$. Describe the quotient. **Sol.** The relation is reflexive, symmetric, and transitive; $[x] = x + \mathbb{Z}$. Every class has exactly one representative in $[0, 1)$, so the quotient is parametrized by that half-open interval.
(fractional part). Map $[x] \mapsto x - [x]$. Integer translation leaves the fractional part unchanged. Two representatives in $[0, 1)$ differing by an integer differ by 0 , proving uniqueness.

Goal	Useful start	Typical finish
$A = B$	Two inclusions	element chase
uniqueness	assume two objects	force equality
nonexistence	suppose object exists	contradiction
$\neg P$	two direct negs	cite both

Relations, cardinality, and proof diagnostics

For a relation $R \subseteq X \times A$, distinguish reflexive, symmetric, antisymmetric, and transitive. Symmetric plus antisymmetric does not mean “neither direction”: if both aRb and bRa occur, $a = b$ is partitioned by the symmetric relation. Reflexive and symmetric, transitive; a total order additionally compares every pair. An equivalence relation is reflexive, symmetric, transitive. A partition is a disjoint union of nonempty sets; a partition defines equivalence by membership in the same cell.
Theorem 2A. Cantor’s diagonal argument shows there is no surjection $\mathbb{N} \rightarrow \mathcal{P}(\mathbb{N})$.

Proof. Given $f: A \rightarrow \mathcal{P}(A)$, set $D = \{a \in A : a \notin f(a)\}$. If $D = f(d)$, then $d \in D \iff d \notin D$, contradiction.
Chinese remainder theorem. If $a \equiv b \pmod{m}$, then a and b have the same remainder modulo m . A partial order is reflexive, antisymmetric, transitive; a total order additionally compares every pair. An equivalence relation is reflexive, symmetric, transitive. A partition is a disjoint union of nonempty sets; a partition defines equivalence by membership in the same cell.
Theorem 2A. Cantor’s diagonal argument shows there is no surjection $\mathbb{N} \rightarrow \mathcal{P}(\mathbb{N})$.

Proof. Given $f: A \rightarrow \mathcal{P}(A)$, set $D = \{a \in A : a \notin f(a)\}$. If $D = f(d)$, then $d \in D \iff d \notin D$, contradiction.

Caution. Disjoint arguments construct an object outside an alleged exhaustive list. Specify that the constructed object belongs to the target set and explain why it differs from every listed entry.

Example 1A. To prove $\sqrt{2}$ irrational, assume $\sqrt{2} = p/q$ in lowest terms. Then $p^2 = 2q^2$, so p is even; writing $p = 2p'$, the lemma “even square implies even integer” follows by contraposition from odd squares being odd.

Chinese remainder theorem. If $a \equiv b \pmod{m}$, then a and b have the same remainder modulo m . A partial order is reflexive, antisymmetric, transitive; a total order additionally compares every pair. An equivalence relation is reflexive, symmetric, transitive. A partition is a disjoint union of nonempty sets; a partition defines equivalence by membership in the same cell.

Proof. If $p \nmid a$, then $\gcd(p, a) = 1$, so Bezout gives $px + ay = 1$. Reducing modulo p gives $ay \equiv 1 \pmod{p}$, hence a is invertible modulo p . Existence of factorization follows by strong induction; the lemma strips matching primes from the uniqueness.

Chinese remainder theorem. If $a \equiv b \pmod{m}$, then a and b have the same remainder modulo m . A partial order is reflexive, antisymmetric, transitive; a total order additionally compares every pair. An equivalence relation is reflexive, symmetric, transitive. A partition is a disjoint union of nonempty sets; a partition defines equivalence by membership in the same cell.

Chinese remainder theorem. If $a \equiv b \pmod{m}$, then a and b have the same remainder modulo m . A partial order is reflexive, antisymmetric, transitive; a total order additionally compares every pair. An equivalence relation is reflexive, symmetric, transitive. A partition is a disjoint union of nonempty sets; a partition defines equivalence by membership in the same cell.

Chinese remainder theorem. If $a \equiv b \pmod{m}$, then a and b have the same remainder modulo m . A partial order is reflexive, antisymmetric, transitive; a total order additionally compares every pair. An equivalence relation is reflexive, symmetric, transitive. A partition is a disjoint union of nonempty sets; a partition defines equivalence by membership in the same cell.

Chinese remainder theorem. If $a \equiv b \pmod{m}$, then a and b have the same remainder modulo m . A partial order is reflexive, antisymmetric, transitive; a total order additionally compares every pair. An equivalence relation is reflexive, symmetric, transitive. A partition is a disjoint union of nonempty sets; a partition defines equivalence by membership in the same cell.

Chinese remainder theorem. If $a \equiv b \pmod{m}$, then a and b have the same remainder modulo m . A partial order is reflexive, antisymmetric, transitive; a total order additionally compares every pair. An equivalence relation is reflexive, symmetric, transitive. A partition is a disjoint union of nonempty sets; a partition defines equivalence by membership in the same cell.

Chinese remainder theorem. If $a \equiv b \pmod{m}$, then a and b have the same remainder modulo m . A partial order is reflexive, antisymmetric, transitive; a total order additionally compares every pair. An equivalence relation is reflexive, symmetric, transitive. A partition is a disjoint union of nonempty sets; a partition defines equivalence by membership in the same cell.

Chinese remainder theorem. If $a \equiv b \pmod{m}$, then a and b have the same remainder modulo m . A partial order is reflexive, antisymmetric, transitive; a total order additionally compares every pair. An equivalence relation is reflexive, symmetric, transitive. A partition is a disjoint union of nonempty sets; a partition defines equivalence by membership in the same cell.

Chinese remainder theorem. If $a \equiv b \pmod{m}$, then a and b have the same remainder modulo m . A partial order is reflexive, antisymmetric, transitive; a total order additionally compares every pair. An equivalence relation is reflexive, symmetric, transitive. A partition is a disjoint union of nonempty sets; a partition defines equivalence by membership in the same cell.

Chinese remainder theorem. If $a \equiv b \pmod{m}$, then a and b have the same remainder modulo m . A partial order is reflexive, antisymmetric, transitive; a total order additionally compares every pair. An equivalence relation is reflexive, symmetric, transitive. A partition is a disjoint union of nonempty sets; a partition defines equivalence by membership in the same cell.

Chinese remainder theorem. If $a \equiv b \pmod{m}$, then a and b have the same remainder modulo m . A partial order is reflexive, antisymmetric, transitive; a total order additionally compares every pair. An equivalence relation is reflexive, symmetric, transitive. A partition is a disjoint union of nonempty sets; a partition defines equivalence by membership in the same cell.

Chinese remainder theorem. If $a \equiv b \pmod{m}$, then a and b have the same remainder modulo m . A partial order is reflexive, antisymmetric, transitive; a total order additionally compares every pair. An equivalence relation is reflexive, symmetric, transitive. A partition is a disjoint union of nonempty sets; a partition defines equivalence by membership in the same cell.

Chinese remainder theorem. If $a \equiv b \pmod{m}$, then a and b have the same remainder modulo m . A partial order is reflexive, antisymmetric, transitive; a total order additionally compares every pair. An equivalence relation is reflexive, symmetric, transitive. A partition is a disjoint union of nonempty sets; a partition defines equivalence by membership in the same cell.

Chinese remainder theorem. If $a \equiv b \pmod{m}$, then a and b have the same remainder modulo m . A partial order is reflexive, antisymmetric, transitive; a total order additionally compares every pair. An equivalence relation is reflexive, symmetric, transitive. A partition is a disjoint union of nonempty sets; a partition defines equivalence by membership in the same cell.

Chinese remainder theorem. If $a \equiv b \pmod{m}$, then a and b have the same remainder modulo m . A partial order is reflexive, antisymmetric, transitive; a total order additionally compares every pair. An equivalence relation is reflexive, symmetric, transitive. A partition is a disjoint union of nonempty sets; a partition defines equivalence by membership in the same cell.

Chinese remainder theorem. If $a \equiv b \pmod{m}$, then a and b have the same remainder modulo m . A partial order is reflexive, antisymmetric, transitive; a total order additionally compares every pair. An equivalence relation is reflexive, symmetric, transitive. A partition is a disjoint union of nonempty sets; a partition defines equivalence by membership in the same cell.

Chinese remainder theorem. If $a \equiv b \pmod{m}$, then a and b have the same remainder modulo m . A partial order is reflexive, antisymmetric, transitive; a total order additionally compares every pair. An equivalence relation is reflexive, symmetric, transitive. A partition is a disjoint union of nonempty sets; a partition defines equivalence by membership in the same cell.

Chinese remainder theorem. If $a \equiv b \pmod{m}$, then a and b have the same remainder modulo m . A partial order is reflexive, antisymmetric, transitive; a total order additionally compares every pair. An equivalence relation is reflexive, symmetric, transitive. A partition is a disjoint union of nonempty sets; a partition defines equivalence by membership in the same cell.

Chinese remainder theorem. If $a \equiv b \pmod{m}$, then a and b have the same remainder modulo m . A partial order is reflexive, antisymmetric, transitive; a total order additionally compares every pair. An equivalence relation is reflexive, symmetric, transitive. A partition is a disjoint union of nonempty sets; a partition defines equivalence by membership in the same cell.

Chinese remainder theorem. If $a \equiv b \pmod{m}$, then a and b have the same remainder modulo m . A partial order is reflexive, antisymmetric, transitive; a total order additionally compares every pair. An equivalence relation is reflexive, symmetric, transitive. A partition is a disjoint union of nonempty sets; a partition defines equivalence by membership in the same cell.

ALG-B dimension formula: $\dim(U+W) = \dim U + \dim W - \dim(U \cap W)$

Proof. Choose a basis $c_1, \dots, c_r$ of $U \cap W$, extend it by $u_1, \dots, u_p$ to U and by $w_1, \dots, w_q$ to W . The concatenated list $u_1, \dots, u_p, w_1, \dots, w_q$ spans $U+W$. If a linear combination vanishes, move the u ’s to the U -side; it lies in $U \cap W$, and independence in U extends base makes all coefficients zero. Count.

For ordered bases $B = (b_1, \dots, b_n)$ and $C = (c_1, \dots, c_m)$, the matrix $[T]_{C \leftarrow B}$ has columns $[T(b_j)]_C$. Thus $[T(v)]_C = [T]_{C \leftarrow B}[v]_B$. $[S \circ T]_{D \leftarrow B} = [S]_{D \leftarrow C}[T]_{C \leftarrow B}$. If $P = [id]_{B \leftarrow C}$, then $[v]_B = P[v]_C$ and the same operator has matrices $A_C = P^{-1}A_BP$. Read arrows before multiplying.

Theorem 6A. For an inner-product space, $\langle u, v \rangle = \langle \|u\| \|v\| \cos \theta \rangle$. For an inner-product space, $\langle u, v \rangle = \langle \|u\| \|v\| \cos \theta \rangle$. For an inner-product space, $\langle u, v \rangle = \langle \|u\| \|v\| \cos \theta \rangle$.

Theorem 6. For $A \in M_n(\mathbb{R})$, the following are equivalent: A is diagonalizable; $\mathbb{R}^n$ has a basis of eigenvectors; the sum of eigenspace dimensions is n ; the minimal polynomial splits into distinct linear factors.

Caution. Pivot columns must be taken from the original matrix. Row operations preserve row space and rank, but generally change column space.

Eigenvalues and diagonalization

Theorem 6. For $A \in M_n(\mathbb{R})$, the following are equivalent: A is diagonalizable; $\mathbb{R}^n$ has a basis of eigenvectors; the sum of eigenspace dimensions is n ; the minimal polynomial splits into distinct linear factors.

Caution. Pivot columns must be taken from the original matrix. Row operations preserve row space and rank, but generally change column space.

Eigenvalues and diagonalization

Theorem 6. For $A \in M_n(\mathbb{R})$, the following are equivalent: A is diagonalizable; $\mathbb{R}^n$ has a basis of eigenvectors; the sum of eigenspace dimensions is n ; the minimal polynomial splits into distinct linear factors.

Caution. Pivot columns must be taken from the original matrix. Row operations preserve row space and rank, but generally change column space.

Eigenvalues and diagonalization

Theorem 6. For $A \in M_n(\mathbb{R})$, the following are equivalent: A is diagonalizable; $\mathbb{R}^n$ has a basis of eigenvectors; the sum of eigenspace dimensions is n ; the minimal polynomial splits into distinct linear factors.

Caution. Pivot columns must be taken from the original matrix. Row operations preserve row space and rank, but generally change column space.

Eigenvalues and diagonalization

Theorem 6. For $A \in M_n(\mathbb{R})$, the following are equivalent: A is diagonalizable; $\mathbb{R}^n$ has a basis of eigenvectors; the sum of eigenspace dimensions is n ; the minimal polynomial splits into distinct linear factors.

Caution. Pivot columns must be taken from the original matrix. Row operations preserve row space and rank, but generally change column space.

Eigenvalues and diagonalization

Theorem 6. For $A \in M_n(\mathbb{R})$, the following are equivalent: A is diagonalizable; $\mathbb{R}^n$ has a basis of eigenvectors; the sum of eigenspace dimensions is n ; the minimal polynomial splits into distinct linear factors.

Caution. Pivot columns must be taken from the original matrix. Row operations preserve row space and rank, but generally change column space.

Eigenvalues and diagonalization

Theorem 6. For $A \in M_n(\mathbb{R})$, the following are equivalent: A is diagonalizable; $\mathbb{R}^n$ has a basis of eigenvectors; the sum of eigenspace dimensions is n ; the minimal polynomial splits into distinct linear factors.

Caution. Pivot columns must be taken from the original matrix. Row operations preserve row space and rank, but generally change column space.

Eigenvalues and diagonalization

Theorem 6. For $A \in M_n(\mathbb{R})$, the following are equivalent: A is diagonalizable; $\mathbb{R}^n$ has a basis of eigenvectors; the sum of eigenspace dimensions is n ; the minimal polynomial splits into distinct linear factors.

Caution. Pivot columns must be taken from the original matrix. Row operations preserve row space and rank, but generally change column space.

Eigenvalues and diagonalization

Theorem 6. For $A \in M_n(\mathbb{R})$, the following are equivalent: A is diagonalizable; $\mathbb{R}^n$ has a basis of eigenvectors; the sum of eigenspace dimensions is n ; the minimal polynomial splits into distinct linear factors.

Caution. Pivot columns must be taken from the original matrix. Row operations preserve row space and rank, but generally change column space.

Eigenvalues and diagonalization

Theorem 6. For $A \in M_n(\mathbb{R})$, the following are equivalent: A is diagonalizable; $\mathbb{R}^n$ has a basis of eigenvectors; the sum of eigenspace dimensions is n ; the minimal polynomial splits into distinct linear factors.

Caution. Pivot columns must be taken from the original matrix. Row operations preserve row space and rank, but generally change column space.

Eigenvalues and diagonalization

Theorem 6. For $A \in M_n(\mathbb{R})$, the following are equivalent: A is diagonalizable; $\mathbb{R}^n$ has a basis of eigenvectors; the sum of eigenspace dimensions is n ; the minimal polynomial splits into distinct linear factors.

Caution. Pivot columns must be taken from the original matrix. Row operations preserve row space and rank, but generally change column space.

Eigenvalues and diagonalization

Theorem 6. For $A \in M_n(\mathbb{R})$, the following are equivalent: A is diagonalizable; $\mathbb{R}^n$ has a basis of eigenvectors; the sum of eigenspace dimensions is n ; the minimal polynomial splits into distinct linear factors.

Caution. Pivot columns must be taken from the original matrix. Row operations preserve row space and rank, but generally change column space.

Eigenvalues and diagonalization

Theorem 6. For $A \in M_n(\mathbb{R})$, the following are equivalent: A is diagonalizable; $\mathbb{R}^n$ has a basis of eigenvectors; the sum of eigenspace dimensions is n ; the minimal polynomial splits into distinct linear factors.

Improper-integral tests

Integral/test	Condition	Conclusion
$\int_a^\infty x^{-p} dx$	$p > 1$	converges
$\int_a^\infty \frac{1}{x^p} dx$	$0 < p \leq 1$	diverges
comparison	$0 \leq f \leq g$, $\int_a^\infty g < \infty$	$\int_a^\infty f < \infty$
limit comparison	$f/g \rightarrow c \in (0, \infty)$	same behavior

w3 [CAL-C] parameter integral. Evaluate $I(a) = \int_0^1 \frac{x-1}{x+a} dx$ for $a > -1$. **Sol.** Differentiate under the integral sign after domination on compact a -intervals: $I'(a) = \int_0^1 x^2 dx = 1/(a+1)$. Since $I(0) = 0$, integration gives $I(a) = \log(a+1)$.

Alt. (Fubini). Use $(x^2-1)/\log x = \int_0^x t^2 dt$ and interchange integrals; absolute integrability on a bounded- t -interval supports the justification.

Sequences and power series

Theorem 8. A power series $\sum_{n \geq 0} a_n (x - c)^n$ has a radius $R \in [0, \infty]$. It converges absolutely for $|x - c| < R$ and diverges for $|x - c| > R$. Endpoints require separate tests. Inside the interval it may be differentiated and integrated termwise without changing R .

(1-x)^{-1} = \sum_{n \geq 0} x^n, e^x = \sum_{n \geq 0} x^n/n!

log(1+x) = \sum_{n \geq 1} (-1)^{n+1} x^n/n.

valid respectively for $|x| < 1$, all x , and $-1 < x \leq 1$ with endpoint analysis. Ratio test: if $|a_{n+1}/a_n| \leq L < 1$, then the series converges absolutely for $L < 1$ and diverges for $L > 1$.

Example 4. $\sum_{n \geq 1} n^{-1}$ diverges although terms tend to 0 . The alternating harmonic series converges by the alternating-series test but not absolutely. Its truncation error after N terms is at most $1/(N+1)$ and has the sign of the first omitted term.

ODE and vector calculus

A first-order linear equation $y' + p(x)y = q(x)$ has integrating factor $\mu = e^{\int p}$ and $(\mu y)' = \mu q$. A separable equation $y' = g(x)h(y)$ permits division by $h(y)$ only after recording equilibrium solutions where $h(y) = 0$. For $y' = \alpha y(1-y)/K$ with $0 < y(0) < K$, separation gives $y(t) = K/(1 + Ce^{-\alpha t})$.

Theorem 8A. If $P = (P_x, P_y)$ is continuously differentiable on a simply connected planar domain and $P_y = Q_x$, then $F = \nabla \phi$ and line integrals depend only on endpoints.

To find ϕ , integrate $P_x \phi_x = P_y \phi_y = \mu q$. A separable equation $y' = g(x)h(y)$ permits division by $h(y)$ only after recording equilibrium solutions where $h(y) = 0$. For $y' = \alpha y(1-y)/K$ with $0 < y(0) < K$, separation gives $y(t) = K/(1 + Ce^{-\alpha t})$.

Example 4. $\sum_{n \geq 1} n^{-1}$ diverges although terms tend to 0 . The alternating harmonic series converges by the alternating-series test but not absolutely. Its truncation error after N terms is at most $1/(N+1)$ and has the sign of the first omitted term.

ODE and vector calculus

A first-order linear equation $y' + p(x)y = q(x)$ has integrating factor $\mu = e^{\int p}$ and $(\mu y)' = \mu q$. A separable equation $y' = g(x)h(y)$ permits division by $h(y)$ only after recording equilibrium solutions where $h(y) = 0$. For $y' = \alpha y(1-y)/K$ with $0 < y(0) < K$, separation gives $y(t) = K/(1 + Ce^{-\alpha t})$.

To find ϕ , integrate $P_x \phi_x = P_y \phi_y = \mu q$. A separable equation $y' = g(x)h(y)$ permits division by $h(y)$ only after recording equilibrium solutions where $h(y) = 0$. For $y' = \alpha y(1-y)/K$ with $0 < y(0) < K$, separation gives $y(t) = K/(1 + Ce^{-\alpha t})$.

Example 4. $\sum_{n \geq 1} n^{-1}$ diverges although terms tend to 0 . The alternating harmonic series converges by the alternating-series test but not absolutely. Its truncation error after N terms is at most $1/(N+1)$ and has the sign of the first omitted term.

ODE and vector calculus

A first-order linear equation $y' + p(x)y = q(x)$ has integrating factor $\mu = e^{\int p}$ and $(\mu y)' = \mu q$. A separable equation $y' = g(x)h(y)$ permits division by $h(y)$ only after recording equilibrium solutions where $h(y) = 0$. For $y' = \alpha y(1-y)/K$ with $0 < y(0) < K$, separation gives $y(t) = K/(1 + Ce^{-\alpha t})$.

To find ϕ , integrate $P_x \phi_x = P_y \phi_y = \mu q$. A separable equation $y' = g(x)h(y)$ permits division by $h(y)$ only after recording equilibrium solutions where $h(y) = 0$. For $y' = \alpha y(1-y)/K$ with $0 < y(0) < K$, separation gives $y(t) = K/(1 + Ce^{-\alpha t})$.

Example 4. $\sum_{n \geq 1} n^{-1}$ diverges although terms tend to 0 . The alternating harmonic series converges by the alternating-series test but not absolutely. Its truncation error after N terms is at most $1/(N+1)$ and has the sign of the first omitted term.

ODE and vector calculus

A first-order linear equation $y' + p(x)y = q(x)$ has integrating factor $\mu = e^{\int p}$ and $(\mu y)' = \mu q$. A separable equation $y' = g(x)h(y)$ permits division by $h(y)$ only after recording equilibrium solutions where $h(y) = 0$. For $y' = \alpha y(1-y)/K$ with $0 < y(0) < K$, separation gives $y(t) = K/(1 + Ce^{-\alpha t})$.

To find ϕ , integrate $P_x \phi_x = P_y \phi_y = \mu q$. A separable equation $y' = g(x)h(y)$ permits division by $h(y)$ only after recording equilibrium solutions where $h(y) = 0$. For $y' = \alpha y(1-y)/K$ with $0 < y(0) < K$, separation gives $y(t) = K/(1 + Ce^{-\alpha t})$.

Example 4. $\sum_{n \geq 1} n^{-1}$ diverges although terms tend to 0 . The alternating harmonic series converges by the alternating-series test but not absolutely. Its truncation error after N terms is at most $1/(N+1)$ and has the sign of the first omitted term.

ODE and vector calculus

A first-order linear equation $y' + p(x)y = q(x)$ has integrating factor $\mu = e^{\int p}$ and $(\mu y)' = \mu q$. A separable equation $y' = g(x)h(y)$ permits division by $h(y)$ only after recording equilibrium solutions where $h(y) = 0$. For $y' = \alpha y(1-y)/K$ with $0 < y(0) < K$, separation gives $y(t) = K/(1 + Ce^{-\alpha t})</$

at most $\delta/2$. Hence $\|f_n - f\|_{\infty} \leq \delta$ for all sufficiently large n . If $f \neq 0$, choose $f_0 = f$ constant, and convergence at one point is already uniform. Compactness is used precisely to make this finite.

Alt. (contradiction by subsequences.) If convergence were not uniform, choose $n_k \uparrow \infty$ and x_k with $|f_{n_k}(x_k) - f(x_k)| \geq \delta$. Then $\|f_{n_k} - f\|_{\infty} \geq \delta$ for all k . But the common Lipschitz bound reduced the error to the pointwise error at x_k , contradiction.

Caution. Pointwise convergence of continuous functions alone does not imply uniform convergence or continuity of the limit: $x^n \rightarrow 0$ on $[0, 1]$ and $x^n \rightarrow 1$ at $x = 1$. The shared modulus of continuity is finite.

PRB-E Probability: conditioning, expectation, and transforms
Events and conditioning

Definition 14. A probability space $(\Omega, \mathcal{F}, \mathbb{P})$, where $\mathcal{F}$ is a sigma-algebra on Ω , $\mathbb{P} : \mathcal{F} \rightarrow [0, 1]$ is countably additive, and $\mathbb{P}(\Omega) = 1$. For $\mathcal{B}(B) = \{A \in \mathcal{F} : A \subseteq B\}$, $\mathbb{P}(A|B) = \mathbb{P}(A \cap B)/\mathbb{P}(B)$. Total probability: $\mathbb{P}(A) = \sum_i \mathbb{P}(A|B_i)\mathbb{P}(B_i)$.

Bayes: $\mathbb{P}(B_j|A) = \frac{\mathbb{P}(A|B_j)\mathbb{P}(B_j)}{\sum_i \mathbb{P}(A|B_i)\mathbb{P}(B_i)}$

The partition (B_i) must be disjoint, exhaustive, and have positive probabilities when conditional terms are Bayes; also requires $\mathbb{P}(A) > 0$. Fairwise independence concerns pairs, not n -tuples. Independence requires the product rule for every finite sub-algebra.

nWS [PRB-E] Bayesian update. A device comes from lines 1 and 2 with probabilities $0, 4$, defect rates are $0.02, .05$. Given a defect, find the probability it came from line 1: $\mathbb{P}(X=1|D) = \frac{0.02 \cdot 0.5}{0.02 \cdot 0.5 + 0.05 \cdot 0.5} = 0.32$. Thus $\mathbb{P}(D|X=1) = 0.02 \cdot 0.5 / 0.32 = 5/8$.

Random variables and expectation

For nonnegative X , Tonelli gives the tail formulas

$$EX = \int_0^\infty \mathbb{P}(X > t) dt,$$
$$EX = \sum_{k=1}^\infty \mathbb{P}(X \geq k) \quad (X \text{ integer-valued}).$$

For finite sums, linearity $\sum_i \mathbb{E}(a_i X_i) = \sum_i a_i EX_i$ needs integrability, not independence. With finite second moments, variance obeys $\text{Var}(X) = \mathbb{E}(X^2) - (EX)^2$ and

$$\text{Var}\left(\sum_i X_i\right) = \sum_i \text{Var}(X_i) + 2 \sum_{i < j} \text{Cov}(X_i, X_j).$$

Example 6. For indicators $I_i = 1_A$, $\mathbb{E} \sum_i I_i = \mathbb{P}(A)$. Counting fixed points of a uniformly random permutation gives expectation $n(1/n) = 1$, despite dependence among the indicators.

For equally likely finite outcomes, $\mathbb{P}(A) = |A|/|\Omega|$ only after specifying the equiprobable sample space. Sampling r objects without replacement from N with M marked objects gives

$$\mathbb{P}(X = k) = \frac{\binom{M}{k} \binom{N-M}{r-k}}{\binom{N}{r}}$$

the feasible range is $\max(0, r - N + K) \leq k \leq \min(r, K)$. With replacement, marked draws are binomial instead.

Method 1.4a. For a Markov model, condition on the first transition using a state that retains the restart probability. Pair their resulting recurrence with boundary data and prove uniqueness, or identify the minimal nonnegative solution.

Example 6A. A simple symmetric random walk on $\{0, 1, \dots, N\}$ stops at 0 or N . If h_i is the chance of reaching N first from i , then $h_i = (h_{i-1} + h_{i+1})/2$, $h_0 = 0$, $h_N = 1$. Second differences vanish, so $h_i = i/N$. For a biased walk, solve $h_i = ph_{i+1} + (1-p)h_{i-1}$.

Caution. Recursion induction needs boundary data and uniqueness. Optional stopping requires hypotheses: bounded T suffices; another standard regime is $ET < \infty$ with a deterministic bound on martingale increments. Uniform integrability of the stopped family is another route.

Markov chains and Poisson processes

A transition matrix P has nonnegative rows summing to one $(P^T \mathbf{1})_i = \mathbb{P}(X_n = j \mid X_0 = i)$. A distribution is stationary when $\pi P = \pi$. Detailed balance $\pi_i P_{ij} = \pi_j P_{ji}$ implies stationarity by summing over i , but stationarity need not imply reversibility.

nWSA [PRB-E] stationary law. For $P = \begin{pmatrix} .8 & .2 \\ .3 & .7 \end{pmatrix}$, find π . **Sol.** Solve $\pi_1 = .8\pi_1 + .3\pi_2$ with $\pi_1 + \pi_2 = 1$. Thus $2\pi_1 = .3\pi_2$, so $\pi = (.6, .4)$.

First-order analysis for a hitting probability h_i gives $h_i = \sum_j P_{ij} h_j$ away from target boundaries. Expected hitting times satisfy $m_i = 1 + \sum_j P_{ij} m_j$. Infinite-state chains require care: a formal solution may not be the desired minimal nonnegative one. For a rate- λ Poisson process, $N(t)$ is Poisson(λt). Increments on disjoint intervals are independent, and waiting times are iid exponential(λ). Conditional on $N(t) = n$, the unordered event times behave as iid uniform points on $[0, t]$; ordered times are their order statistics.

Example 6B. Superposing independent Poisson processes of rates λ_1, λ_2 increases rate λ . Given an arrival, its source label is 1 with probability $\lambda_1/(\lambda_1 + \lambda_2)$, independently across arrivals.

Entropy and information inequalities

For finite-alphabet variables, $H(X) = -\sum_i p(x) \log p(x)$ with $0 \log 0 = 0$. Conditional entropy averages $H(X \mid Y = y)$; mutual information

$$I(X; Y) = \sum_{x,y} p(x, y) \log \frac{p(x, y)}{p(x)p(y)} = H(X, Y) - H(X \mid Y) \geq 0.$$

It vanishes exactly when X, Y are independent. Chain rule: $H(X, Y, Z) = H(X) + H(Y \mid X) + H(Z \mid X, Y)$. Conditioning cannot increase entropy.

Theorem 14C. Gibbs' inequality: $D(p||q) = \sum_i p_i \log(p_i/q_i) \geq 0$, with $D = \infty$ if some $p_i > q_i$. For p, q probability distributions, $D(p||q) \leq \log 2$ iff $p_i \leq 2q_i$ for all i . **Proof.** Observe $\log u \leq u - 1$ with $u = q(x)/p(x)$ over $p(x) > 0$: $-D(p||q) \leq \sum_i p_i (q_i/p_i - 1) \leq 0$.

Example 6C. A fair bit has entropy $\log 2$. If it passes through n independent symmetric channels with probability ε that they flip, then $H(Y) = \log 2$ and $H(Y \mid X) = h(\varepsilon)$, so $I(X; Y) = \log 2 - h(\varepsilon)$. Information falls to zero at $\varepsilon = 1/2$.

Caution. Differential entropy may be negative and changes under coordinate scaling. Mutual information and relative entropy retain invariant operational meaning under bijective reparametrization.

PRB-E Probability: conditioning, distributions and inequalities
Transforms and named laws

Law	support	mean	variance
Bernoulli(p)	$\{0, 1\}$	p	$p(1-p)$
Binomial(n, p)	$\{0, \dots, n\}$	np	$np(1-p)$
Poisson(λ)	$\mathbb{N}$	λ	λ
Geometric(p)	$1, 2, \dots$	$1/p$	$(1-p)/p^2$
Exponential(λ)	$x \geq 0$	$1/\lambda$	$1/\lambda^2$
Normal(μ, σ^2)	$\mathbb{R}$	μ	σ^2

Parameters: binomial $n \in \mathbb{N}_0$, Bernoulli/binomial $p \in [0, 1]$, geometric $p \in (0, 1]$, Poisson/exponential $\lambda > 0$, and nondegenerate normal $\sigma^2 > 0$. The moment generating function $M_X(t) = \mathbb{E}(e^{tX})$, when finite on a neighborhood of 0, determines the law and satisfies $M_X^{(k)}(0) = \mathbb{E}(X^k)$. Independence gives $M_{X,Y} = M_X M_Y$. Probability generating functions $G_X(s) = \mathbb{E}(s^X)$ are often safer for nonnegative integer-valued variables.

Theorem 15. Markov: for $X \geq 0$ and $a > 0$, $\mathbb{P}(X \geq a) \leq \mathbb{E}X/a$. Chebyshev: if $\text{Var}(X) > 0$, then $\mathbb{P}(|X - \mathbb{E}X| \geq a) \leq \text{Var}(X)/a^2$.

Proof. Since $X \geq a1_{\{X \geq a\}}$, take expectations for Markov. Apply Markov to $(X - \mathbb{E}X)^2$ for Chebyshev.

nWS [PRB-E2] Let $X_1, \dots, X_n$ be independent with mean μ and variance $\sigma^2 > 0$. For $\varepsilon > 0$, bound $\mathbb{P}(\bar{X}_n - \mu \geq \varepsilon)$ and identify the limiting conclusion.

Caution. Differential entropy may be negative and changes under coordinate scaling. Mutual information and relative entropy retain invariant operational meaning under bijective reparametrization.

PRB-E2 Probability: conditioning, distributions and inequalities
Transforms and named laws

Law support mean variance

Bernoulli(p) $\{0, 1\}$ p $p(1-p)$

Binomial(n, p) $\{0, \dots, n\}$ np $np(1-p)$

Poisson(λ) $\mathbb{N}$ λ λ

Geometric(p) $1, 2, \dots$ $1/p$ $(1-p)/p^2$

Exponential(λ) $x \geq 0$ $1/\lambda$ $1/\lambda^2$

Normal(μ, σ^2) $\mathbb{R}$ μ σ^2

Parameters: binomial $n \in \mathbb{N}_0$, Bernoulli/binomial $p \in [0, 1]$, geometric $p \in (0, 1]$, Poisson/exponential $\lambda > 0$, and nondegenerate normal $\sigma^2 > 0$. The moment generating function $M_X(t) = \mathbb{E}(e^{tX})$, when finite on a neighborhood of 0, determines the law and satisfies $M_X^{(k)}(0) = \mathbb{E}(X^k)$. Independence gives $M_{X,Y} = M_X M_Y$. Probability generating functions $G_X(s) = \mathbb{E}(s^X)$ are often safer for nonnegative integer-valued variables.

Theorem 15. Markov: for $X \geq 0$ and $a > 0$, $\mathbb{P}(X \geq a) \leq \mathbb{E}X/a$. Chebyshev: if $\text{Var}(X) > 0$, then $\mathbb{P}(|X - \mathbb{E}X| \geq a) \leq \text{Var}(X)/a^2$.

Proof. Since $X \geq a1_{\{X \geq a\}}$, take expectations for Markov. Apply Markov to $(X - \mathbb{E}X)^2$ for Chebyshev.

nWS [PRB-E2] Let $X_1, \dots, X_n$ be independent with mean μ and variance $\sigma^2 > 0$. For $\varepsilon > 0$, bound $\mathbb{P}(\bar{X}_n - \mu \geq \varepsilon)$ and identify the limiting conclusion.

Caution. Differential entropy may be negative and changes under coordinate scaling. Mutual information and relative entropy retain invariant operational meaning under bijective reparametrization.

PRB-E2 Probability: conditioning, distributions and inequalities
Transforms and named laws

Law support mean variance

Bernoulli(p) $\{0, 1\}$ p $p(1-p)$

Binomial(n, p) $\{0, \dots, n\}$ np $np(1-p)$

Poisson(λ) $\mathbb{N}$ λ λ

Geometric(p) $1, 2, \dots$ $1/p$ $(1-p)/p^2$

Exponential(λ) $x \geq 0$ $1/\lambda$ $1/\lambda^2$

Normal(μ, σ^2) $\mathbb{R}$ μ σ^2

Parameters: binomial $n \in \mathbb{N}_0$, Bernoulli/binomial $p \in [0, 1]$, geometric $p \in (0, 1]$, Poisson/exponential $\lambda > 0$, and nondegenerate normal $\sigma^2 > 0$. The moment generating function $M_X(t) = \mathbb{E}(e^{tX})$, when finite on a neighborhood of 0, determines the law and satisfies $M_X^{(k)}(0) = \mathbb{E}(X^k)$. Independence gives $M_{X,Y} = M_X M_Y$. Probability generating functions $G_X(s) = \mathbb{E}(s^X)$ are often safer for nonnegative integer-valued variables.

Theorem 15. Markov: for $X \geq 0$ and $a > 0$, $\mathbb{P}(X \geq a) \leq \mathbb{E}X/a$. Chebyshev: if $\text{Var}(X) > 0$, then $\mathbb{P}(|X - \mathbb{E}X| \geq a) \leq \text{Var}(X)/a^2$.

Proof. Since $X \geq a1_{\{X \geq a\}}$, take expectations for Markov. Apply Markov to $(X - \mathbb{E}X)^2$ for Chebyshev.

nWS [PRB-E2] Let $X_1, \dots, X_n$ be independent with mean μ and variance $\sigma^2 > 0$. For $\varepsilon > 0$, bound $\mathbb{P}(\bar{X}_n - \mu \geq \varepsilon)$ and identify the limiting conclusion.

Caution. Differential entropy may be negative and changes under coordinate scaling. Mutual information and relative entropy retain invariant operational meaning under bijective reparametrization.

PRB-E2 Probability: conditioning, distributions and inequalities
Transforms and named laws

Law support mean variance

Bernoulli(p) $\{0, 1\}$ p $p(1-p)$

Binomial(n, p) $\{0, \dots, n\}$ np $np(1-p)$

Poisson(λ) $\mathbb{N}$ λ λ

Geometric(p) $1, 2, \dots$ $1/p$ $(1-p)/p^2$

Exponential(λ) $x \geq 0$ $1/\lambda$ $1/\lambda^2$

Normal(μ, σ^2) $\mathbb{R}$ μ σ^2

Parameters: binomial $n \in \mathbb{N}_0$, Bernoulli/binomial $p \in [0, 1]$, geometric $p \in (0, 1]$, Poisson/exponential $\lambda > 0$, and nondegenerate normal $\sigma^2 > 0$. The moment generating function $M_X(t) = \mathbb{E}(e^{tX})$, when finite on a neighborhood of 0, determines the law and satisfies $M_X^{(k)}(0) = \mathbb{E}(X^k)$. Independence gives $M_{X,Y} = M_X M_Y$. Probability generating functions $G_X(s) = \mathbb{E}(s^X)$ are often safer for nonnegative integer-valued variables.

Theorem 15. Markov: for $X \geq 0$ and $a > 0$, $\mathbb{P}(X \geq a) \leq \mathbb{E}X/a$. Chebyshev: if $\text{Var}(X) > 0$, then $\mathbb{P}(|X - \mathbb{E}X| \geq a) \leq \text{Var}(X)/a^2$.

Proof. Since $X \geq a1_{\{X \geq a\}}$, take expectations for Markov. Apply Markov to $(X - \mathbb{E}X)^2$ for Chebyshev.

nWS [PRB-E2] Let $X_1, \dots, X_n$ be independent with mean μ and variance $\sigma^2 > 0$. For $\varepsilon > 0$, bound $\mathbb{P}(\bar{X}_n - \mu \geq \varepsilon)$ and identify the limiting conclusion.

Caution. Differential entropy may be negative and changes under coordinate scaling. Mutual information and relative entropy retain invariant operational meaning under bijective reparametrization.

PRB-E2 Probability: conditioning, distributions and inequalities
Transforms and named laws

Law support mean variance

Bernoulli(p) $\{0, 1\}$ p $p(1-p)$

Binomial(n, p) $\{0, \dots, n\}$ np $np(1-p)$

Poisson(λ) $\mathbb{N}$ λ λ

Geometric(p) $1, 2, \dots$ $1/p$ $(1-p)/p^2$

Exponential(λ) $x \geq 0$ $1/\lambda$ $1/\lambda^2$

Normal(μ, σ^2) $\mathbb{R}$ μ σ^2

Parameters: binomial $n \in \mathbb{N}_0$, Bernoulli/binomial $p \in [0, 1]$, geometric $p \in (0, 1]$, Poisson/exponential $\lambda > 0$, and nondegenerate normal $\sigma^2 > 0$. The moment generating function $M_X(t) = \mathbb{E}(e^{tX})$, when finite on a neighborhood of 0, determines the law and satisfies $M_X^{(k)}(0) = \mathbb{E}(X^k)$. Independence gives $M_{X,Y} = M_X M_Y$. Probability generating functions $G_X(s) = \mathbb{E}(s^X)$ are often safer for nonnegative integer-valued variables.

Theorem 15. Markov: for $X \geq 0$ and $a > 0$, $\mathbb{P}(X \geq a) \leq \mathbb{E}X/a$. Chebyshev: if $\text{Var}(X) > 0$, then $\mathbb{P}(|X - \mathbb{E}X| \geq a) \leq \text{Var}(X)/a^2$.

Proof. Since $X \geq a1_{\{X \geq a\}}$, take expectations for Markov. Apply Markov to $(X - \mathbb{E}X)^2$ for Chebyshev.

nWS [PRB-E2] Let $X_1, \dots, X_n$ be independent with mean μ and variance $\sigma^2 > 0$. For $\varepsilon > 0$, bound $\mathbb{P}(\bar{X}_n - \mu \geq \varepsilon)$ and identify the limiting conclusion.

Caution. Differential entropy may be negative and changes under coordinate scaling. Mutual information and relative entropy retain invariant operational meaning under bijective reparametrization.

PRB-E2 Probability: conditioning, distributions and inequalities
Transforms and named laws

Law support mean variance

Bernoulli(p) $\{0, 1\}$ p $p(1-p)$

Binomial(n, p) $\{0, \dots, n\}$ np $np(1-p)$

Poisson(λ) $\mathbb{N}$ λ λ

Geometric(p) $1, 2, \dots$ $1/p$ $(1-p)/p^2$

Exponential(λ) $x \geq 0$ $1/\lambda$ $1/\lambda^2$

Normal(μ, σ^2) $\mathbb{R}$ μ σ^2

Parameters: binomial $n \in \mathbb{N}_0$, Bernoulli/binomial $p \in [0, 1]$, geometric $p \in (0, 1]$, Poisson/exponential $\lambda > 0$, and nondegenerate normal $\sigma^2 > 0$. The moment generating function $M_X(t) = \mathbb{E}(e^{tX})$, when finite on a neighborhood of 0, determines the law and satisfies $M_X^{(k)}(0) = \mathbb{E}(X^k)$. Independence gives $M_{X,Y} = M_X M_Y$. Probability generating functions $G_X(s) = \mathbb{E}(s^X)$ are often safer for nonnegative integer-valued variables.

Theorem 15. Markov: for $X \geq 0$ and $a > 0$, $\mathbb{P}(X \geq a) \leq \mathbb{E}X/a$. Chebyshev: if $\text{Var}(X) > 0$, then $\mathbb{P}(|X - \mathbb{E}X| \geq a) \leq \text{Var}(X)/a^2$.

Proof. Since $X \geq a1_{\{X \geq a\}}$, take expectations for Markov. Apply Markov to $(X - \mathbb{E}X)^2$ for Chebyshev.

nWS [PRB-E2] Let $X_1, \dots, X_n$ be independent with mean μ and variance $\sigma^2 > 0$. For $\varepsilon > 0$, bound $\mathbb{P}(\bar{X}_n - \mu \geq \varepsilon)$ and identify the limiting conclusion.

Caution. Differential entropy may be negative and changes under coordinate scaling. Mutual information and relative entropy retain invariant operational meaning under bijective reparametrization.

PRB-E2 Probability: conditioning, distributions and inequalities
Transforms and named laws

Law support mean variance

Bernoulli(p) $\{0, 1\}$ p $p(1-p)$

Binomial(n, p) $\{0, \dots, n\}$ np $np(1-p)$

Poisson(λ) $\mathbb{N}$ λ λ

Geometric(p) $1, 2, \dots$ $1/p$ $(1-p)/p^2$

Exponential(λ) $x \geq 0$ $1/\lambda$ $1/\lambda^2$

Normal(μ, σ^2) $\mathbb{R}$ μ σ^2

Parameters: binomial $n \in \mathbb{N}_0$, Bernoulli/binomial $p \in [0, 1]$, geometric $p \in (0, 1]$, Poisson/exponential $\lambda > 0$, and nondegenerate normal $\sigma^2 > 0$. The moment generating function $M_X(t) = \mathbb{E}(e^{tX})$, when finite on a neighborhood of 0, determines the law and satisfies $M_X^{(k)}(0) = \mathbb{E}(X^k)$. Independence gives $M_{X,Y} = M_X M_Y$. Probability generating functions $G_X(s) = \mathbb{E}(s^X)$ are often safer for nonnegative integer-valued variables.

Theorem 15. Markov: for $X \geq 0$ and $a > 0$, $\mathbb{P}(X \geq a) \leq \mathbb{E}X/a$. Chebyshev: if $\text{Var}(X) > 0$, then $\mathbb{P}(|X - \mathbb{E}X| \geq a) \leq \text{Var}(X)/a^2$.

Proof. Since $X \geq a1_{\{X \geq a\}}$, take expectations for Markov. Apply Markov to $(X - \mathbb{E}X)^2$ for Chebyshev.

nWS [PRB-E2] Let $X_1, \dots, X_n$ be independent with mean μ and variance $\sigma^2 > 0$. For $\varepsilon > 0$, bound $\mathbb{P}(\bar{X}_n - \mu \geq \varepsilon)$ and identify the limiting conclusion.

Caution. Differential entropy may be negative and changes under coordinate scaling. Mutual information and relative entropy retain invariant operational meaning under bijective reparametrization.

PRB-E2 Probability: conditioning, distributions and inequalities
Transforms and named laws

Law support mean variance

Bernoulli(p) $\{0, 1\}$ p $p(1-p)$

Binomial(n, p) $\{0, \dots, n\}$ np $np(1-p)$

Poisson(λ) $\mathbb{N}$ λ λ

Geometric(p) $1, 2, \dots$ $1/p$ $(1-p)/p^2$

Exponential(λ) $x \geq 0$ $1/\lambda$ $1/\lambda^2$

Normal(μ, σ^2) $\mathbb{R}$ μ σ^2

Parameters: binomial $n \in \mathbb{N}_0$, Bernoulli/binomial $p \in [0, 1]$, geometric $p \in (0, 1]$, Poisson/exponential $\lambda > 0$, and nondegenerate normal $\sigma^2 > 0$. The moment generating function $M_X(t) = \mathbb{E}(e^{tX})$, when finite on a neighborhood of 0, determines the law and satisfies $M_X^{(k)}(0) = \mathbb{E}(X^k)$. Independence gives $M_{X,Y} = M_X M_Y$. Probability generating functions $G_X(s) = \mathbb{E}(s^X)$ are often safer for nonnegative integer-valued variables.

Theorem 15. Markov: for $X \geq 0$ and $a > 0$, $\mathbb{P}(X \geq a) \leq \mathbb{E}X/a$. Chebyshev: if $\text{Var}(X) > 0$, then $\mathbb{P}(|X - \mathbb{E}X| \geq a) \leq \text{Var}(X)/a^2$.

Proof. Since $X \geq a1_{\{X \geq a\}}$, take expectations for Markov. Apply Markov to $(X - \mathbb{E}X)^2$ for Chebyshev.

nWS [PRB-E2] Let $X_1, \dots, X_n$ be independent with mean μ and variance $\sigma^2 > 0$. For $\varepsilon > 0$, bound $\mathbb{P}(\bar{X}_n - \mu \geq \varepsilon)$ and identify the limiting conclusion.

Caution. Differential entropy may be negative and changes under coordinate scaling. Mutual information and relative entropy retain invariant operational meaning under bijective reparametrization.

PRB-E2 Probability: conditioning, distributions and inequalities
Transforms and named laws

Law support mean variance

Bernoulli(p) $\{0, 1\}$ p $p(1-p)$

Binomial(n, p) $\{0, \dots, n\}$ np $np(1-p)$

Poisson(λ) $\mathbb{N}$ λ λ

Geometric(p) $1, 2, \dots$ $1/p$ $(1-p)/p^2$

Exponential(λ) $x \geq 0$ $1/\lambda$ $1/\lambda^2$

Normal(μ, σ^2) $\mathbb{R}$ μ σ^2

Parameters: binomial $n \in \mathbb{N}_0$, Bernoulli/binomial $p \in [0, 1]$, geometric $p \in (0, 1]$, Poisson/exponential $\lambda > 0$, and nondegenerate normal $\sigma^2 > 0$. The moment generating function $M_X(t) = \mathbb{E}(e^{tX})$, when finite on a neighborhood of 0, determines the law and satisfies $M_X^{(k)}(0) = \mathbb{E}(X^k)$. Independence gives $M_{X,Y} = M_X M_Y$. Probability generating functions $G_X(s) = \mathbb{E}(s^X)$ are often safer for nonnegative integer-valued variables.

Theorem 15. Markov: for $X \geq 0$ and $a > 0$, $\mathbb{P}(X \geq a) \leq \mathbb{E}X/a$. Chebyshev: if $\text{Var}(X) > 0$, then $\mathbb{P}(|X - \mathbb{E}X| \geq a) \leq \text{Var}(X)/a^2$.

Proof. Since $X \geq a1_{\{X \geq a\}}$, take expectations for Markov. Apply Markov to $(X - \mathbb{E}X)^2$ for Chebyshev.

nWS [PRB-E2] Let $X_1, \dots, X_n$ be independent with mean μ and variance $\sigma^2 > 0$. For $\varepsilon > 0$, bound $\mathbb{P}(\bar{X}_n - \mu \geq \varepsilon)$ and identify the limiting conclusion.

Caution. Differential entropy may be negative and changes under coordinate scaling. Mutual information and relative entropy retain invariant operational meaning under bijective reparametrization.

PRB-E2 Probability: conditioning, distributions and inequalities
Transforms and named laws

Law support mean variance

Bernoulli(p) $\{0, 1\}$ p $p(1-p)$

Binomial(n, p) $\{0, \dots, n\}$ np $np(1-p)$

Poisson(λ) $\mathbb{N}$ λ λ

Geometric(p) $1, 2, \dots$ $1/p$ $(1-p)/p^2$

Exponential(λ) $x \geq 0$ $1/\lambda$ $1/\lambda^2$

Normal(μ, σ^2) $\mathbb{R}$ μ σ^2

Parameters: binomial $n \in \mathbb{N}_0$, Bernoulli/binomial $p \in [0, 1]$, geometric $p \in (0, 1]$, Poisson/exponential $\lambda > 0$, and nondegenerate normal $\sigma^2 > 0$. The moment generating function $M_X(t) = \mathbb{E}(e^{tX})$, when finite on a neighborhood of 0, determines the law and satisfies $M_X^{(k)}(0) = \mathbb{E}(X^k)$. Independence gives $M_{X,Y} = M_X M_Y$. Probability generating functions $G_X(s) = \mathbb{E}(s^X)$ are often safer for nonnegative integer-valued variables.

Theorem 15. Markov: for $X \geq 0$ and $a > 0$, $\mathbb{P}(X \geq a) \leq \mathbb{E}X/a$. Chebyshev: if $\text{Var}(X) > 0$, then $\mathbb{P}(|X - \mathbb{E}X| \geq a) \leq \text{Var}(X)/a^2$.

Proof. Since $X \geq a1_{\{X \geq a\}}$, take expectations for Markov. Apply Markov to $(X - \mathbb{E}X)^2$ for Chebyshev.

nWS [PRB-E2] Let $X_1, \dots, X_n$ be independent with mean μ and variance $\sigma^2 > 0$. For $\varepsilon > 0$, bound $\mathbb{P}(\bar{X}_n - \mu \geq \varepsilon)$ and identify the limiting conclusion.

Caution. Differential entropy may be negative and changes under coordinate scaling. Mutual information and relative entropy retain invariant operational meaning under bijective reparametrization.

PRB-E2 Probability: conditioning, distributions and inequalities
Transforms and named laws

Law support mean variance

Bernoulli(p) $\{0, 1\}$ p $p(1-p)$

Binomial(n, p) $\{0, \dots, n\}$ np $np(1-p)$

Poisson(